

Writing Paper's Background and Related Work Sections

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Content adapted from:

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Outline

- Background
- Related work
- Exercise

Typical Paper Organization

- Title, along with names, affiliations, email addresses of authors
- Abstract
- Introduction
- Background
- Related work
- Proposed approach
- Evaluation
 - Experimental design
 - Results
 - Discussions
- Threats to validity (optional)
- Conclusions

Background

- Provides background knowledge to understand the problem and your method
 - e.g., if your method is using Convolutional Neural Networks (CNN) then you discuss CNN in the background section
- Background often provides
 - A summary of the problem being addressed
 - Previous method from which your solution relies on
 - Definitions on concepts needed to understand your method
 - Existing theories on which your solution is based
- Background is different from related work
 - Related work discusses existing approaches to addressing the problem

Exercise

- Suppose that you write a background section describing the Internet Protocol (IP). Please write one different version from the following taken from Wikipedia (https://en.wikipedia.org/wiki/Internet_Protocol):

“The Internet Protocol (IP) is the principal communications protocol in the Internet protocol suite for relaying datagrams across network boundaries. Its routing function enables internetworking, and essentially establishes the Internet.

IP has the task of delivering packets from the source host to the destination host solely based on the IP addresses in the packet headers. For this purpose, IP defines packet structures that encapsulate the data to be delivered. It also defines addressing methods that are used to label the datagram with source and destination information”

Please work in groups of two each in 20 minutes

Related work

- Presents alternative approaches to the same or closely related problems/topics
- Compares and contrasts existing approaches, related them to your approach
 - Give credit to existing approaches while showing how your approach is different from them
- Provides strengths and weaknesses of related work
- Length: dependent the space available and relevance of the related work
 - Conference paper: $\frac{1}{2}$ - $1 \frac{1}{2}$ pages
 - Journal paper: 1 - 4 pages

Where to put the related work section

- After the introduction/example section
 - Pros: immediately clear out reviewers' wonder on how the work differs from previous work
 - Cons: hard to let readers to know what you are talking about before showing the approach details
- Before the conclusion section
 - Pros: now reviewers' know what your approach is about
 - Cons: reviewers keep wondering how the work differs from previous work till this point
 - But for very closely related work, you should have pointed out the differences in the introduction section

Related work – tips 1

- Related work can be organized into subsections or grouped into several categories
- Common words to use
 - “describe”, “present”, “propose”, “show”, “point out”, “suggest”, “recommend”, “find”, “argue”
 - Don’t repeat one word all the time
- Don’t simply list related work **without** RELATING to your own work!
 - keywords to use: whereas, in contrast, but, however, ...
 - “excuses” to use: “does not require specs”, “focus on different problems”, “complement with each other”, ...
 - you can describe several similar related approaches together and compare them at once with yours

Related work – tips 2

- Don't make unjustified unobvious criticisms on related work if you don't have experimental results to back you up
 - But you can cite others' experiments to back you up
- Don't overclaim your work without justification
- Don't intentionally leave out your own very related previous papers (reviewers can find them out easily)
 - maybe even need to mention them in Introduction section and explain why the new work is different
 - reviewers often try to identify a marginal/incremental paper or a “least publishable unit (LPU)”

Related work – tips 3

- Be modest in discussing strengths and weaknesses of previous approaches
- Make sure your reviews of related work are correct
 - Read related work in detail
 - Read other authors' reviews of related work
 - Check the limitations mentioned in related work

Related work – tips 4

■ Simple ways to find related work

- ❑ Use Google Scholar
- ❑ Find a related paper
- ❑ Check papers citing it

- ❑ Read the related section of papers

A Constrained Regression Technique for COCOMO Calibration



A constrained regression technique for COCOMO calibration

[V Nguyen](#), [B Steece](#), [B Boehm](#) - Proceedings of the Second ACM-IEEE ..., 2008 - dl.acm.org

Building cost estimation models is often considered a search problem in which the solver should return an optimal solution satisfying an objective function. This solution also needs to meet certain constraints. For example, a solution for the estimates coefficients of COCOMO models must be non-negative. In this research, we introduce a constrained regression technique that uses objective functions and constraints to estimate the coefficients of the COCOMO models. To access the performance of the proposed technique, we run a cross ...



Cited by 72

Related articles

All 9 versions

The truth: credit is not like money

- Warmly acknowledge people who have helped you
- Be generous to the competition

“In his inspiring paper [Foo98] Foogole shows.... We develop his foundation in the following ways...”

- Acknowledge weaknesses in your approach

Giving credit to others does not diminish the credit you get from your paper

Source: “[How to write a great research paper - Seven simple suggestions](#)” by Simon Peyton Jones

Plagiarism

- People often commit plagiarism when writing the background and related work
- Plagiarism is the act of reusing an existing published material without proper acknowledgement
- Using your own published work is also plagiarism – called self-plagiarism
- If plagiarism is found in your submission, your paper is rejected
 - If not, your reputation is damaged

Preventing plagiarism - 1

- Citing sources properly!
- General rules: do not write more than 3 consecutive words same as the original source
- If you copy sentences/phrases verbatim, quote them and cite sources
- Replacing words with synonyms is not enough
- Some submission tools like EDAS provide a similarity index showing the similarity between your paper and others

Preventing plagiarism - 2

- Read the following quotation and writings from four students.
- Which papers are plagiarized? Why?

Being educated means being skillful with language—able to control language instead of being controlled by it, confident that you can speak or write effectively instead of feeling terrified. When successful people explain how they rose to the top, they often emphasize their skills as communicators ... Writing, private or public, ... is really about you, about the richness of your life lived in language, about the fullness of your participation in your community and in your culture, about the effectiveness of your efforts to achieve change. The person attuned to the infinite creativity of language leads a richer life. So can you. (Gardner & Barefoot, 2014, p. 175)

Student paper 1	Student paper 2	Student paper 3	Student paper 4
Skill with language, both spoken and written, is one characteristic of an educated person. Many people attribute their success to their skills as communicators	Educated people are skillful communicators. They use their knowledge of language, both spoken and written, to help them in their personal and professional lives (Gardner & Barefoot, 2014)	The term <i>educated</i> , as defined by Gardner and Barefoot (2014), means efficiency in using the communication skills of speaking and writing to foster growth and change in both the public and private sectors of life	One can either control language or be controlled by it. Educated people continually strive to improve their skills as communicators so that they can control language and become more successful at it

From Mary Renck Jalongo, Olivia N. Saracho, “Writing for Publication”, Springer

Exercise

- Work with your paper partner
- Identify a topic of your research
- Use tools like Google Scholar, Semantic Scholar to identify related papers to the topic
- Store papers' citations in **Semantic Scholar**, Mendeley, Zotero or another similar tool